Linear Dynamical System

Calculate $A^m X_0$ ✓

$$x_n = 0.9x_{n-1} + 0.2y_{n-1}$$
$$y_n = 0.1x_{n-1} + 0.8y_{n-1}$$

$$X_n = A^m X_0$$

$$\begin{pmatrix} x_n \\ y_n \end{pmatrix} = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix} \begin{pmatrix} x_{n-1} \\ y_{n-1} \end{pmatrix}$$

$$X_n = AX_{n-1}$$

$$X_1 = AX_0$$
$$X_2 = AX_1 = A(AX_0) = A^2X_0$$
$$X_3 = AX_2 = A(A^2X_0) = A^3X_0$$
$$\vdots$$

Method-1: $A^n X_0 = (P D^n P^{-1}) X_0$

$A = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}$, $X_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\det(A - \lambda I) = 0$ ← Characteristic eq.

$\lambda^2 - 1.7\lambda + 0.7 = 0$

$\lambda = 1, 0.7$

$D = \begin{pmatrix} 1 & 0 \\ 0 & 0.7 \end{pmatrix}$

$\lambda = 1 \quad (A - \lambda I) X = 0$

$\begin{pmatrix} 0.9-1 & 0.2 & 0 \\ 0.1 & 0.8-1 & 0 \end{pmatrix}$

$= \begin{pmatrix} -0.1 & 0.2 & 0 \\ 0.1 & -0.2 & 0 \end{pmatrix}$

$-0.1x + 0.2y = 0$

$x = 2y$

$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2y \\ y \end{pmatrix} = y \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

$\lambda = 1, \quad 0.7$

$\downarrow \qquad \downarrow$

$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$D^n = \begin{pmatrix} 1^n & 0 \\ 0 & 0.7^n \end{pmatrix}$

$P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}$

$A^n X_0 = (P D^n P^{-1}) X_0$

$= \frac{-1}{3} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1^n & 0 \\ 0 & 0.7^n \end{pmatrix} \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$

$A^n X_0$

Method - II:

$A^n X_0$

$\{v_1, v_2\}$ → Basis $\mathbb{R}^2$

$X_0 = c_1 v_1 + c_2 v_2$

$A^n X_0 = c_1 A^n v_1 + c_2 A^n v_2$

$= c_1 (\lambda_1^n v_1) + c_2 \lambda_2^n v_2$

$$A^n X_0 = c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2$$

$Ax = \lambda x$

A^2 ?

$A(Ax) = A(\lambda x)$

$A^2 x = \lambda(Ax) = \lambda(\lambda x)$

$= \lambda^2 x$

$A^2 x = \lambda^2 x$

$$A^n x_0 = c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2 + c_3 \lambda_3^n v_3 + \cdots + c_k \lambda_k^n v_k .$$

$$A = \begin{pmatrix} 1 & 3 \\ 0 & 5 \end{pmatrix}, \quad X_0 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

$A^n X_0 = ?$

$\lambda = 1, 5$

$$A^n X_0 = c_1 \lambda_1^n v_1 + c_2 \lambda_2^n v_2$$

$$\boxed{A^n X_0 = c_1 1^n \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 5^n \begin{pmatrix} 3 \\ 4 \end{pmatrix}}$$

$\lambda = 1$

$(A - 1I)X = 0$

$$\begin{pmatrix} 0 & 3 & 0 \\ 0 & 4 & 0 \end{pmatrix}$$

$y = 0$

$$\begin{pmatrix} x \\ y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$\lambda = 5$

$(A - 5I)X = 0$

$$\begin{pmatrix} -4 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$-4x + 3y = 0$

$x = \frac{3}{4} y$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{3}{4} y \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$y \begin{pmatrix} 3/4 \\ 1 \end{pmatrix}$

$$X_0 = \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \quad v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$c_1 v_1 + c_2 v_2 = X_0$$

$$\left(\begin{array}{cc|c} 1 & 3 & 1 \\ 0 & 4 & 5 \end{array}\right)$$

$$4c_2 = 5 \qquad c_1 + 3\left(\frac{5}{4}\right) = 1$$

$$\boxed{c_2 = \frac{5}{4}} \qquad c_1 = 1 - \frac{15}{4}$$

$$\boxed{c_1 = \frac{-11}{4}}$$

$$A^n X_0 = \frac{-11}{4}(1)^n \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{5}{4}(5)^n \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$